🧬 Assessment 1 — Omics Presentation

Universidad Pública de Navarra

Master in Biomedical Engineering

Module: “Advanced Bioinformatics”

Release date: Tuesday, 12 February 2026
Slide upload: *TBD, 23:59*
In-class presentation: *TBD*
Weight: 20 % of the module mark

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1. Purpose

Your first assignment is a **10–15 minute oral presentation** built on one or more review articles provided by the instructors. The aim is **not** to master every omics technology, but to explore this field from your engineering curiosity—understand the problem, the proposed technology, and its potential impact on biomedical engineering or personalised medicine.

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2. Article selection & signup

1. **Pick your article(s):** browse the official article list. You may cover **one** review or combine up to **three** if they share a coherent storyline.
2. **Reserve your slot:** write your name in the shared sheet. Capacity per article is limited—first come, first served.

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3. Presentation structure (10–15 min + Q&A)

1. **Personal intro:** why you chose the topic / what you hoped to learn.
2. **Problem & motivation:** biomedical need addressed by the review.
3. **Technology overview:** explain key omics concepts in your own words, using analogies if needed.
4. **Experimental/computational workflow:** include at least one **original diagram** (BioRender, PowerPoint, Figma) showing samples, data flow, and analysis steps.
5. **Main findings / proposals:** what makes the article novel or disruptive.
6. **Impact on biomedical engineering:** opportunities, limitations, ethical considerations.
7. **Open questions / future steps:** what you would investigate next, uncertainties that remain.
8. **References and figure credits.**

> Be transparent about sections you did not fully understand—describe how you would clarify them. Honesty and curiosity are part of the evaluation.

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4. Preparing the talk

- **Break the reading** into blocks (context, methods, case studies) and summarise in your own words.
- **Rehearse with a timer** once or twice to stay within 10–15 minutes.
- **Use guiding questions:** What problem does it solve? What data/sensors does it need? Which steps limit adoption?
- **Design for clarity:** avoid text-heavy slides, highlight key figures, include your own diagram.
- **Engage the audience:** finish with 1–2 questions for classmates or the instructor.

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5. Evaluation rubric

Criterion	Description	Weight
Presentation quality	Storytelling, clarity, pacing, audience-friendly language.	20 %
Visual support	Slide structure, original diagram(s), proper references.	20 %
Understanding	Ability to explain the article's concepts, limitations, and contributions.	20 %
Q&A performance	Handling questions, openness to feedback, teamwork in answers.	20 %
Participation	Ask questions to your peers, respect for timing, balanced roles.	20 %

> You are not expected to be omics experts; we value your capacity to learn, interpret, and communicate complex ideas credibly.

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6. Resources

- Reference managers: Zotero, Mendeley
- Visual tools: BioRender, PowerPoint, Canva, Figma
- Pitch templates / omics glossary: available on ADI

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7. Pre-presentation checklist

- ☐ Topic reserved in the shared sheet
- ☐ Slides uploaded before the deadline
- ☐ Original workflow diagram included
- ☐ Rehearsal completed (10–15 min)
- ☐ Questions prepared for the Q&A segment

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